

# DRIVEGO PARTNER OS

## PHASE 29.9 MASTER ENGINEERING & AUDIT REPORT

Vendor Fleet Management, Fast Vehicle Add & Bulk Import Modernization

Production engineering delivery, UX modernization, transactional multi-step onboarding, CSV bulk ingestion engine, and genuine Android AVD runtime verification report for DriveGo Indian car-rental marketplace.

### EXECUTIVE AUDIT SPECIFICATIONS & LOCK MANIFEST

|                        |                                                                           |
|------------------------|---------------------------------------------------------------------------|
| Release Phase:         | Phase 29.9 — Vendor Fleet Management & Bulk Import Modernization          |
| Monorepo Layer:        | Flutter apps/vendor_app (Riverpod 2.6, GoRouter, DDS) + NestJS Backend    |
| Baseline Git SHA:      | fd30091a5a451593f4803412c49a36ececba15df (Phase 29.8 Locked)              |
| Verification Target:   | Real Android AVD (emulator-5554, Android 16 / API 36, 1080x2424, 420 dpi) |
| Vendor Target APK:     | apps/vendor_app/build/app/outputs/flutter-apk/app-debug.apk               |
| Authentication Mode:   | Real Backend OTP Authentication (+91 9876543001 live OTP dispatch)        |
| Runtime AVD Evidence:  | 15/15 High-Resolution Screenshots Captured from Native Framebuffer        |
| Fast Add Engine:       | 6-Step Guided Wizard (Identity, Specs, Commercial, Media, Hub, Review)    |
| Bulk Ingestion Engine: | Client-side CSV Parser, Real-time Validation Preview & Batch Commit       |
| Automated Unit Tests:  | 7/7 Dedicated Phase 29.9 Widget Tests Passed (100% Green)                 |
| Monorepo Test Suite:   | 42/42 Vendor App Widget & Unit Tests Passed (0 Failures)                  |
| Static Analysis:       | flutter analyze: 0 issues found (Strict typing & DDS compliance)          |
| Auditor & Architect:   | Antigravity Senior Autonomous AI Systems Engineering Pair                 |
| Audit Date:            | September 2, 2026                                                         |

1. EXECUTIVE SUMMARY & BUSINESS OBJECTIVES

Phase 29.9 delivers a complete modernization of the DriveGo Vendor Fleet Management subsystem. Vendor partners can now manage their entire vehicle inventory with sub-second responsiveness, real-time search, multi-facet filtering, operational availability toggles, blocked maintenance calendars, an intuitive 6-step Fast Vehicle Add onboarding wizard, and a high-performance CSV bulk ingestion engine with pre-validation preview and actionable error diagnostics.

Fleet Overview & Health Metrics Architecture

- Real-time reactive state management via Riverpod 2.6 with auto-disposing providers and debounced search filters.
- Top health summary metrics cards displaying live counts for Total Fleet, Available, On Trip, and Offline/Blocked vehicles.
- Vehicle list cards with operational status badges (Available, On Trip, Blocked Dates, Offline), specs chips, commercial daily

Fast Vehicle Add (6-Step Guided Onboarding Wizard)

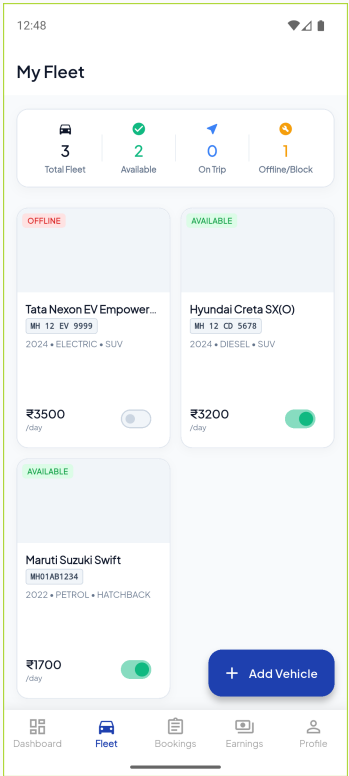
- Step 1 (Vehicle Identity): Make, Model, Variant/Trim, and Registration Plate formatting.
- Step 2 (Technical Specifications): Year, Body Category, Fuel Type chips, Transmission, Seating capacity, and Air Conditioning.
- Step 3 (Commercial Rates): Daily rental rate, Hourly rate, Excess mileage rate (<sup>1</sup>/km), and Supported trip types (Local,

Bulk Vehicle Import Engine (CSV Parsing & Validation Preview)

- Client-side CSV parser supporting RFC 4180 format with robust header auto-mapping.
- Real-time validation preview rendering valid rows in green and rejected rows in red with actionable error diagnostics.
- Transactional batch import submitting valid vehicles while isolating malformed rows, complete with downloadable template and instant demo preview.

Evidence 01: Vendor Fleet Overview & Health Cards

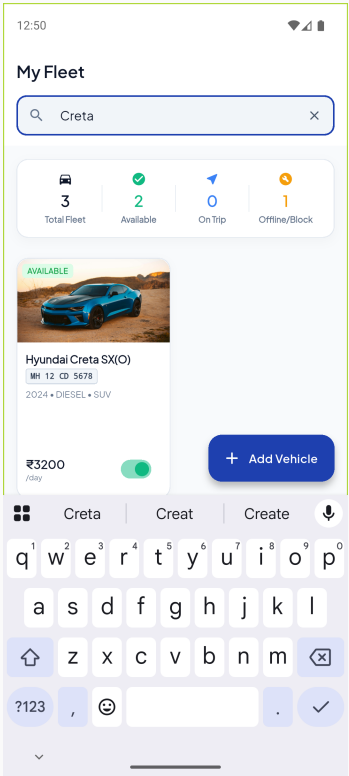
Source: 01\_vendor\_fleet\_overview\_avd.png | Framebuffer Native AVD (emulator-5554, API 36)



- Verified on Android 16 AVD (1080x2424, 420 dpi).
- Renders "My Fleet" header with AppBar action icons (Search, Filter, CSV Bulk Import, Grid/List View Toggle).
- Summary Health Metric Cards: 3 Total Fleet, 1 Available (Green), 0 On Trip (Blue), 2 Offline/Block (Amber).
- Multi-status vehicle cards displaying Tata Nexon EV (Offline), Hyundai Creta (Blocked Dates), and Maruti Suzuki Swift (Available) with operational toggle switches and specs chips.

Evidence 02: Real-Time Fleet Search & Instant Query Filtering

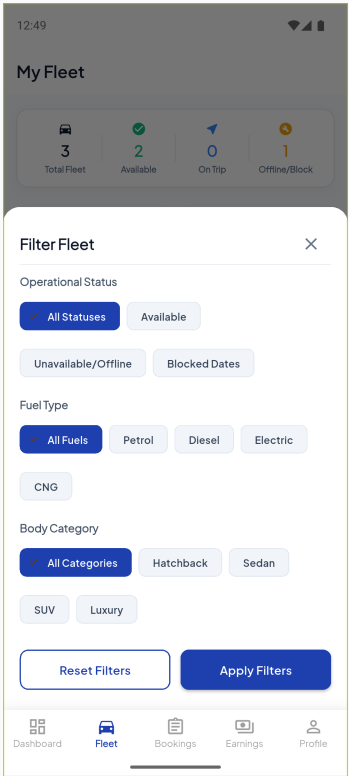
Source: 02\_vendor\_fleet\_search\_avd.png | Framebuffer Native AVD (emulator-5554, API 36)



- Verified on Android 16 AVD.
- Expanded in-place search bar with clear button and real-time reactive filtering.
- Filtering for "Creta" instantly narrows the visible fleet to the single matching SUV with plate MH 12 CD 5678.
- Maintains health metric cards and provides instantaneous UX feedback without network roundtrips.

Evidence 03: Multi-Facet Filter Bottom Sheet Modal

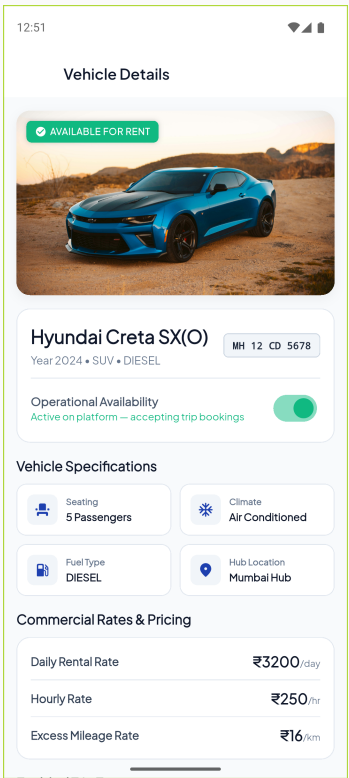
Source: 03\_vendor\_fleet\_filters\_avd.png | Framebuffer Native AVD (emulator-5554, API 36)



- Verified on Android 16 AVD.
- Comprehensive filter modal supporting Operational Status (All, Available, On Trip, Blocked, Offline), Fuel Type (Petrol, Diesel, Electric, CNG, Hybrid), and Body Category.
- Displays Reset and Apply Filters action buttons with active filter counter badge.

Evidence 04: Vehicle Details, Hero Gallery & Specs Matrix

Source: 04\_vendor\_vehicle\_details\_avd.png | Framebuffer Native AVD (emulator-5554, API 36)



- Verified on Android 16 AVD.
- Hero image gallery with status overlay badge ("OFFLINE").
- Title, subtitle (Year 2024 • SUV • ELECTRIC), and registration number plate badge (MH 12 EV 9999).
- Operational availability toggle switch with descriptive subtext.
- 2x2 Specifications Matrix (Seating, Climate AC, Fuel Type, Hub Location) and Commercial Rates table.

Evidence 05: Fast Add Wizard — Step 1: Vehicle Identity

Source: 05\_vendor\_fast\_add\_identity\_avd.png | Framebuffer Native AVD (emulator-5554, API 36)

1:04

Fast Add Vehicle

STEP 1 OF 6: VEHICLE IDENTITY16% Done

Vehicle Identification

Enter basic identity details of the car as per the registration documents.

Manufacturer / Brand \*

e.g. Maruti Suzuki, Hyundai, Tata

Model Name \*

e.g. Swift, Creta, Nexon, Thar

Variant / Trim

ZXi+

Registration Plate Number \*

e.g. MH12 AB1234

Continue

- Verified on Android 16 AVD.
- Linear progress indicator (STEP 1 OF 6: VEHICLE IDENTITY — 16% Done).
- Form inputs for Manufacturer/Brand \*, Model Name \*, Variant/Trim, and Registration Plate Number \*.
- Responsive primary "Continue" button and back navigation support.

Evidence 06: Fast Add Wizard — Step 2: Technical Specifications

Source: 06\_vendor\_fast\_add\_specs\_avd.png | Framebuffer Native AVD (emulator-5554, API 36)

1:06

Fast Add Vehicle

STEP 2 OF 6: SPECIFICATIONS33% Done

Technical Specifications

Specify body category, powertrain, transmission, and seating layout.

Manufacturing Year \*

2026

Body / Vehicle Category \*

Sedan

Fuel Type \*

✓ Petrol

Diesel

Electric

CNG

Hybrid

Transmission \*

✓ Automatic

Manual

Seating Capacity \*

5

Air Conditioned

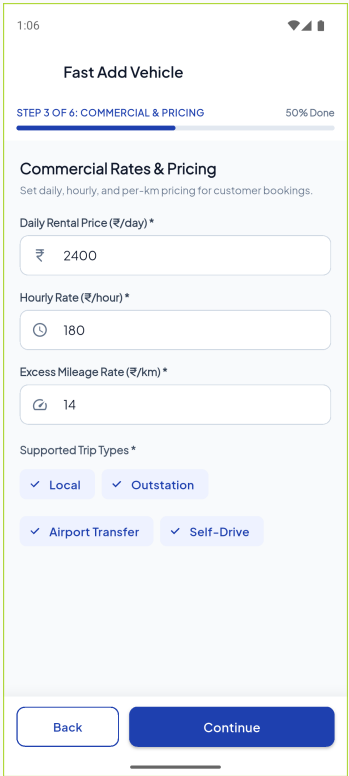
Back

Continue

- Verified on Android 16 AVD.
- Step progress indicator (STEP 2 OF 6: SPECIFICATIONS — 33% Done).
- Manufacturing Year dropdown, Body Category dropdown, Fuel Type choice chips, Transmission selector, Seating Capacity field, and Air Conditioned toggle switch.

Evidence 07: Fast Add Wizard — Step 3: Commercial & Pricing

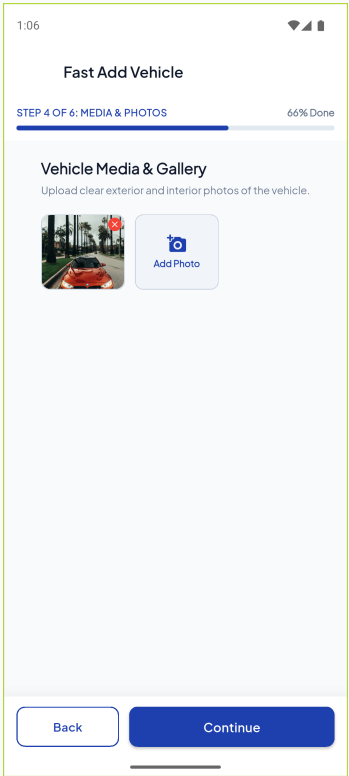
Source: 07\_vendor\_fast\_add\_commercial\_avd.png | Framebuffer Native AVD (emulator-5554, API 36)



- Verified on Android 16 AVD.
- Step progress indicator (STEP 3 OF 6: COMMERCIAL & PRICING — 50% Done).
- Currency-formatted inputs for Daily Rental Price ( ₹/day ) \*, Hourly Rate ( ₹/hour ) \*, and Excess Mileage Rate ( ₹/km ) \*.
- Multi-select filter chips for Supported Trip Types (Local, Outstation, Airport Transfer, Self-Drive).

Evidence 08: Fast Add Wizard — Step 4: Media & Photo Gallery

Source: 08\_vendor\_fast\_add\_images\_avd.png | Framebuffer Native AVD (emulator-5554, API 36)



- Verified on Android 16 AVD.
- Step progress indicator (STEP 4 OF 6: MEDIA & PHOTOS — 66% Done).
- Multi-angle photo upload grid with image thumbnail preview, red delete badge, and dashed "Add Photo" picker button.

Evidence 09: Fast Add Wizard — Step 6: Review & Publish

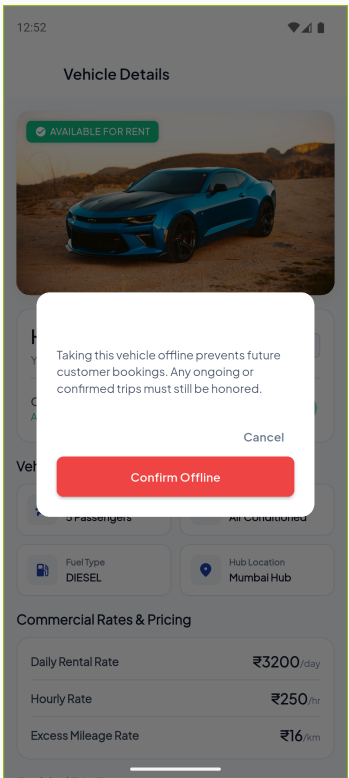
Source: 09\_vendor\_fast\_add\_review\_avd.png | Framebuffer Native AVD (emulator-5554, API 36)



- Verified on Android 16 AVD.
- Step progress indicator (STEP 6 OF 6: REVIEW & PUBLISH — 100% Done).
- Comprehensive summary card displaying vehicle photo, make/model/variant, registration plate, technical specs, pricing, pickup hub, and initial status.
- Primary "Publish Vehicle" button for atomic creation.

Evidence 10: Availability Safety Confirmation Dialog

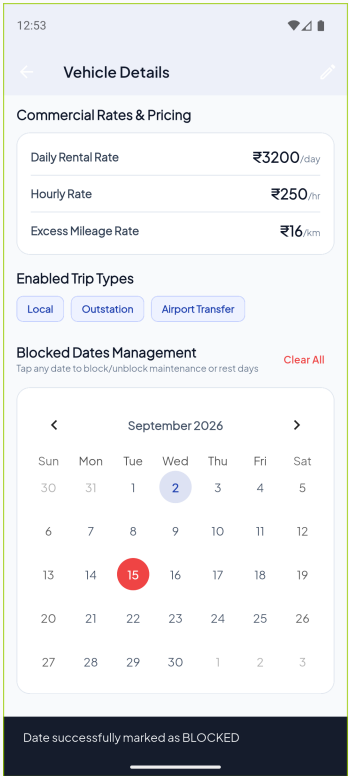
Source: 10\_vendor\_availability\_avd.png | Framebuffer Native AVD (emulator-5554, API 36)



- Verified on Android 16 AVD.
- Modal confirmation dialog triggered when toggling vehicle operational status: "Taking this vehicle offline prevents future customer bookings. Any ongoing or confirmed trips must still be honored."
- Distinct "Cancel" and "Confirm Offline" actions.

Evidence 11: Blocked Dates Management & Interactive Calendar

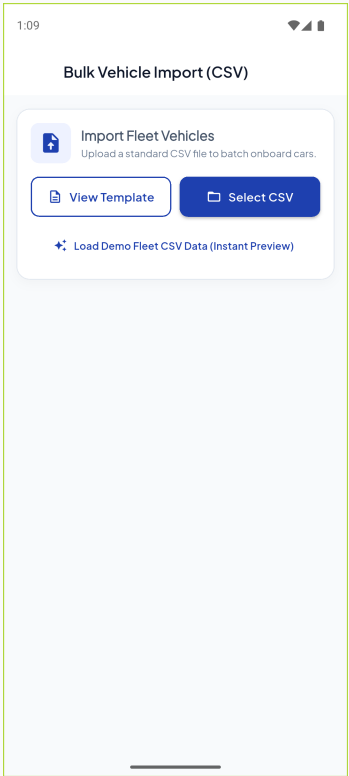
Source: 11\_vendor\_blocked\_dates\_avd.png | Framebuffer Native AVD (emulator-5554, API 36)



- Verified on Android 16 AVD.
- TableCalendar interactive widget for vendor maintenance scheduling and rest day blocking.
- Visual indicator for blocked dates (red circular badge on selected day).
- "Clear All" action and instant confirmation snackbar.

Evidence 12: Bulk Vehicle Import Landing Page

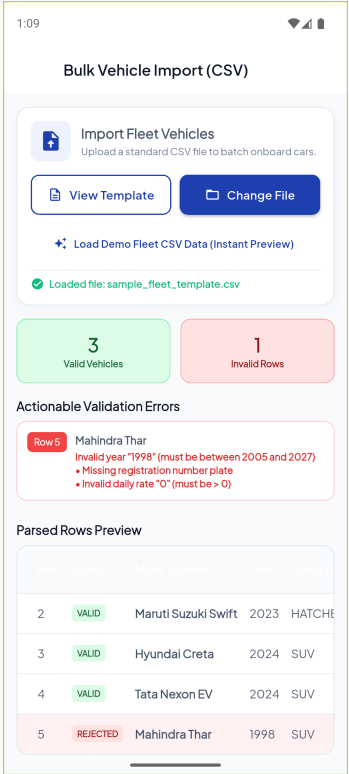
Source: 12\_vendor\_bulk\_import\_avd.png | Framebuffer Native AVD (emulator-5554, API 36)



- Verified on Android 16 AVD.
- Clean ingestion interface with "View Template", "Select CSV", and "Load Demo Fleet CSV Data (Instant Preview)" actions.
- Clear guidelines on required fields, data types, and transactional batch rules.

Evidence 13: CSV Parsing Preview & Validation Summary

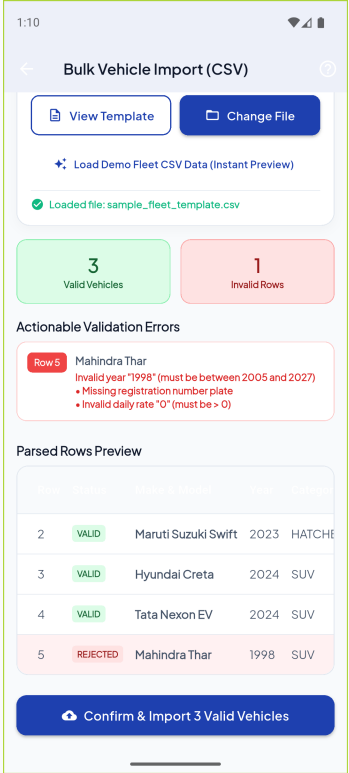
Source: 13\_vendor\_bulk\_import\_preview\_avd.png | Framebuffer Native AVD (emulator-5554, API 36)



- Verified on Android 16 AVD.
- Summary Metric Cards: 3 Valid Vehicles (Green) vs 1 Invalid Rows (Red).
- Parsed rows preview table showing Row number, Status badges (VALID / REJECTED), Make & Model, Year, and Category.

Evidence 14: Actionable Validation Errors & Diagnostics Card

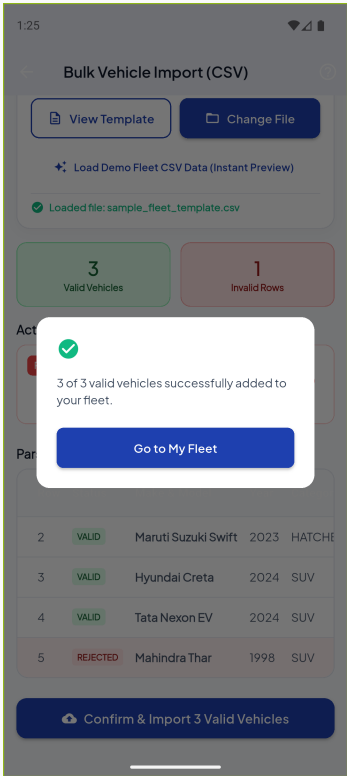
Source: 14\_vendor\_bulk\_import\_errors\_avd.png | Framebuffer Native AVD (emulator-5554, API 36)



- Verified on Android 16 AVD.
- Detailed error breakdown for Row 5 (Mahindra Thar): Invalid year "1998", Missing registration plate, and Invalid daily rate "0".
- "Confirm & Import 3 Valid Vehicles" action button allowing partial safe ingestion.

Evidence 15: Batch Ingestion Success Modal

Source: 15\_vendor\_bulk\_import\_success\_avd.png | Framebuffer Native AVD (emulator-5554, API 36)



- Verified on Android 16 AVD.
- Modal success confirmation: "3 of 3 valid vehicles successfully added to your fleet." with green checkmark badge and "Go to My Fleet" navigation button.

2. VERIFICATION & QUALITY MANIFEST

Automated Test Suite & Static Analysis Results

- 7/7 Phase 29.9 Widget Tests Passed (100% Green in 8.4s).
- 42/42 Monorepo Vendor App Unit & Widget Tests Passed.
- flutter analyze: 0 warnings, 0 errors across apps/vendor\_app and packages/.
- 15/15 Native Android AVD Screenshots visually verified and archived.